

Ben Rosario

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EDUCATION

University of California, Berkeley

Bachelor of Arts in Cognitive Science

Berkeley, CA

Expected May 2028

Sierra College

Associate of Science in Computer Science & Associate of Science in Mathematics

Rocklin, CA

May 2026

Graduated with honors

- **Relevant Coursework:** Data Structures, Linear Algebra, Multivariable Calculus, Discrete Mathematics, Differential Equations, Symbolic Logic, Statistics

PROJECTS

Course Scheduling Assistant | *Python, FastAPI, OpenAI API, FAISS*

Nov. 2025 – Present

- Built and deployed a retrieval-augmented generation (RAG) assistant with a FastAPI backend and Discord frontend on Railway, adopted by 40+ students for course planning
- Implemented hybrid retrieval combining FAISS vector search and IDF-weighted lexical matching to resolve ambiguous course queries, including Linear Algebra versus College Algebra
- Automated hourly Playwright extraction of JSON API responses for approximately 2,000 course records; refreshed the search index without restarting the API
- Managed per-user conversation context with bounded history and expiration; added API rate limits and SQLite analytics to track usage
- Built pytest regression tests for retrieval, conversation state, and data processing; configured GitHub Actions to run tests across Python 3.12–3.14

Mog Scan | *React, TypeScript, FastAPI, MediaPipe, SQLite*

Sept. 2026 – Present

- Built a React/TypeScript face-scanning app integrating MediaPipe tracking with FastAPI model inference; checked pose, lighting, and blur before scoring selected frames and generating downloadable result cards
- Developed anonymous leaderboards, posts, comments, and voting with FastAPI and SQLite; used database transactions and vote revisions to prevent duplicate posts and conflicting vote updates
- Tested API behavior and simultaneous voting/deletion with pytest; verified multi-user publishing, session persistence, and mobile layouts down to 320px with Playwright

EXPERIENCE

Student Body President

Aug. 2025 – May 2026

Associated Students of Sierra College

Rocklin, CA

- Elected to represent 21,000 students as chief executive of student government; oversaw the passage of 63 bills during the academic year
- Directed a \$342,000 annual operating budget and oversaw \$800,000 in reserves
- Presented student positions on federal financial aid and minority-serving institution funding to a congressional representative at the U.S. Capitol

Student Trustee

June 2025 – May 2026

Sierra Joint Community College District Board of Trustees

Rocklin, CA

- Served as the sole student representative on the district Board of Trustees, reviewing district policy and casting advisory votes
- Reviewed contracts and cast advisory votes for science building and student housing projects with total project values of \$84M and \$98M, respectively

Student Tutor

Aug. 2024 – May 2026

Sierra College

Rocklin, CA

- Tutored approximately 15 students per week in Calculus I–III, Java, C, C++, and Data Structures, adapting explanations to each student's background and course level

TECHNICAL SKILLS

Languages: Python, TypeScript, C, C++, Java, Swift, SQL

Libraries & Frameworks: React, FastAPI, MediaPipe, OpenAI SDK, FAISS, Pydantic, discord.py, Playwright, pytest

Tools & Platforms: Git, GitHub Actions, Docker, Railway, SQLite